PRACTICE QUESTIONS JOINT DISCRETE RANDOM VARIABLES

EXERCISE – 3.3

JOINT PROBABILITY DISTRIBUTIONS

1. Determine the values of *c* so that the following functions represent joint probability distributions of the random variables *X* and *Y* :

$$f(x,y) = cxy \quad for\ x = 1,2,3 : y = 1,2,3$$

$(Ans : 1/36)$

2. Determine the values of *c* so that the following functions represent joint probability distributions of the random variables *X* and *Y* :

$$f(x,y) = c|x-y| \quad for\ x = -2,0,2 : y = -2,3$$

$(Ans : 1/15)$

3. From a sack of fruit containing 3 oranges, 2 apples, and 3 bananas, a random sample of 4 pieces of fruit is selected. If *X* is the number of oranges and *Y* is the number of apples in the sample, find
(a) the joint probability distribution of *X* and *Y* ;
(b) $P(x+y \le 2)$ $(Ans : 1/2)$

4. If the joint probability distribution of *X* and *Y* is given by

$$f(x,y) = \frac{x+y}{30} \quad for\ x = 0,1,2,3 : y = 0,1,2$$

Find
(a) $P(X \le 2, Y = 1)$
(b) $P(X > 2, Y \le 1)$
(c) $P(X > Y)$
(d) $P(X + Y = 4)$

$(Ans : 1/5, 7/30, 3/5, 4/15)$

5. Let *X* denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let *Y* denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

$f(x,y)$		x = 1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

(a)Evaluate the marginal distribution of *X*.
(b) Evaluate the marginal distribution of *Y* .
(c) Find *P*(*Y* = 3 | *X* = 2).
$(Ans : 0.2857)$

x	1	2	3
$g(x)$	0.10	0.35	0.55

y	1	2	3
$h(y)$	0.20	0.50	0.30

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6. Suppose that *X* and *Y* have the following joint probability distribution:

		x	
$f(x,y)$		2	4
	1	0.10	0.15
y	3	0.20	0.30
	5	0.10	0.15

(a)Find the marginal distribution of *X* and Y.
(b) *Find* $E(X+Y)$, $E(X-Y)$, $E(XY)$, $E(X)$, $E(Y)$ (c) $COV(X,Y)$ (d) $COV(X,Y)$

Ans :

x	2	4
$g(x)$	0.40	0.60

y	1	3	5
$h(y)$	0.25	0.50	0.25

7. Let *X* denote the: number of heads and *Y* the number of heads minus the number of tails when 3 coins are tossed. Find the joint probability distribution of *X* and *Y*. *also find* $E(X)$, $E(Y)$, $E(XY)$, $COV(X,Y)$, $Corr(X,Y)$

Ans :

		x			
	$f(x,y)$	0	1	2	3
	-3	$1/8$			
y	-1		$3/8$		
	1			$3/8$	
	3				$1/8$

8. Consider an experiment that consists of 2 rolls of a balanced die. If *X* is the number of 4s and *Y* is the number of 5s obtained in the 2 rolls of the die, find
(a) the joint probability distribution of X and Y; (b) $P(2x+y<3)$
$(c)\, E(X)$, $E(Y)$, $E(XY)$, $COV(X,Y)$, $Corr(X,Y)$
$(Ans: 11/12)$

9. Three cards are drawn without replacement from the 12 face cards (jacks, queens, and kings) of an ordinary deck of 52 playing cards. Let *X* be the number of kings selected and *Y* the number of jacks. Find
(a) the joint probability distribution of *X* and *Y* ;
(b) $P(x+y\geq 2)$ (c) $Corr(X,Y)$
$(Ans: 42/55)$

		x			
	$f(x,y)$	0	1	2	3
	0	$\frac{1}{55}$	$\frac{6}{55}$	$\frac{6}{55}$	$\frac{1}{55}$
y	1	$\frac{6}{55}$	$\frac{16}{55}$	$\frac{6}{55}$	0
	2	$\frac{6}{55}$	$\frac{6}{55}$	0	0
	3	$\frac{1}{55}$	0	0	0

10. A coin is tossed twice. Let *Z* denote the number of heads on the first toss and *W* the total number of heads on the 2 tosses. If the coin is unbalanced and a head has a 40% chance of occurring, find
(a) the joint probability distribution of *W* and *Z*;
(b) the marginal distribution of *W* and Z
(c) the probability that at least 1 head occurs.

(*Ans* :

		w		
	$f(w,z)$	0	1	2
z	0	0.36	0.24	
	1		0.24	0.16

w	0	1	2
$g(w)$	0.36	0.48	0.16

z	0	1
$h(z)$	0.60	0.40

$(Ans: P(W\geq 1)=064)$

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